

Shotgun engineering fires up mammalian metabolic design

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By testing random combinations of genes at scale, shotgun genetic engineering enables discovery of new productive metabolic pathways in mammalian cells, including solutions for synthesizing two essential amino acids.

Engineering complex, multigene programs in mammalian cells is among the most challenging problems in synthetic biology. It requires selecting the appropriate genes, tuning their relative expression levels, and directing proteins to the appropriate subcellular compartments. The resulting combinatorial space is often too vast to explore experimentally. Writing in *Nature Biotechnology*, Trolle et al.¹ sidestep this problem with shotgun genetic engineering (SGE), a pooled strategy that avoids rational design and instead harnesses selection to discover complex genetic programs (Fig. 1). SGE floods cells with small, barcoded genetic modules, allowing each cell to express a distinct pathway at random; after phenotypic selection, barcode sequencing identifies productive module combinations. Remarkably, the authors use SGE to endow mammalian cells with the capacity to synthesize two essential amino acids.

Metabolic pathway engineering has traditionally focused on fast-growing microbes such as bacteria and yeast, where design–build–

test–learn cycles can be completed within days². In mammalian cells, design–build–test–learn cycles are far more time-consuming. Slower proliferation rates extend the time required for engineered cells to recover and expand to numbers sufficient for downstream assays. Furthermore, pathway construction may be constrained by inefficient delivery of large genetic constructs into cells. Both challenges compound as pathway complexity increases because the number of possible multigene configurations expands combinatorially while cell delivery becomes progressively less efficient³. Indeed, recent methods for combinatorial screening in mammalian cells have enabled the exploration of vast design spaces, but only for relatively compact multigene constructs (<10 kb)⁴. Computational metabolic models provide another possible route to identify promising perturbations, but models trained on wild-type cells are poorly equipped to design pathways assembled from foreign or synthetic components⁵. A practical strategy for experimental discovery of complex pathways in mammalian cells is still lacking.

SGE addresses these challenges through a key methodological advance. Rather than constructing and permuting a single, large multigene vector, it creates a pooled library of single-gene cassettes, each containing a unique set of genetic parts linked to a DNA barcode. The library is delivered through high multiplicity-of-infection lentiviral transduction, enabling each cell to integrate multiple cassettes at random genomic sites and assemble different pathways from the

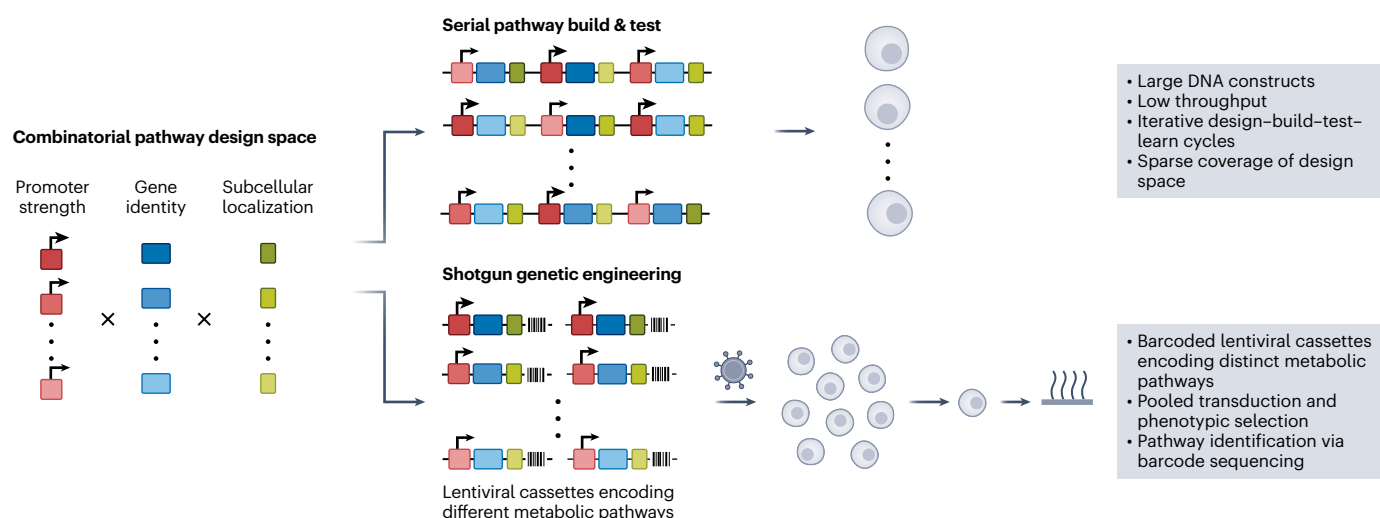


Fig. 1 | Shotgun genetic engineering streamlines the exploration of vast synthetic pathway design spaces. Productive synthetic pathways can arise from complex design spaces comprising combinations of genetic parts including promoters, gene variants and subcellular localization tags (left). In conventional synthetic pathway engineering (top), large DNA constructs are designed and

delivered serially, limiting coverage of pathway design space and requiring repeated design–build–test–learn iterations. In SGE (bottom), a pooled library of small, barcoded gene cassettes is delivered at high copy numbers so that individual cells assemble a distinct pathway at random. Functional pathways are subsequently identified by phenotypic selection and barcode sequencing.

same pool. Variation in both cassette identity and copy number allows the cell population to sample both different part combinations and different relative expression levels. Phenotypic selection is then applied to find functional needles in the combinatorial haystack, and sequencing reveals the cassette combination underlying the phenotype of interest. In this way, SGE alleviates the difficult task of assembling complex multigene vectors, replacing it with a single pooled transduction that searches diverse design compositions and expression states in parallel.

To put SGE to the test, the authors tackled branched-chain amino acid biosynthesis, a metabolic pathway absent in mammals⁵. First, they demonstrated valine biosynthesis in Chinese hamster ovary cells, a workhorse of the biopharmaceutical industry. The SGE library comprised genetic constructs containing different enzyme-coding genes, promoters of variable strengths, and subcellular localization tags; selection in low-valine medium recovered clones with higher metabolic activity than in parental cell lines. Notably, the uncovered clones grew at approximately 80% of their wild-type growth rate in valine-free medium, doubling every 1.1 days, compared with 3.8 days exhibited by the same group's cells engineered with a rationally designed pathway previously⁶. More strikingly, SGE identified a bacterial pathway that enabled Chinese hamster ovary cells to synthesize isoleucine, installing this metabolic capability in mammalian cells for the first time. Across the top-performing designs, a notable pattern emerged: enzymes with mitochondrial localization were highly enriched, revealing that subcellular localization is a critical design variable. As the authors note, this feature – which has been largely overlooked by rational approaches – may reflect the endosymbiotic origin of mitochondria and echoes the compartmentalization of analogous pathways in fungal mitochondria and plant chloroplasts. As a final demonstration, the authors installed valine synthesis in Jurkat cells, a human T cell line, showing that the approach can apply to different cell types. Barcode data were subsequently used to train a machine-learning classifier that accurately predicted function and independently rediscovered the same design rules⁷.

SGE has considerable potential to make inaccessible multigene design spaces experimentally searchable, and several straightforward extensions will undoubtedly broaden its reach in the future. Pairing SGE with engineered biosensors would enable selection for phenotypes that lack a simple growth-based readout, while coupling barcode recovery to single-cell transcriptomics could further reveal how individual genetic programs reshape global cell phenotypes. A pragmatic advance would be to combine SGE with new methods for pooled screening of reconstructed, contiguous multigene programs⁴. This approach would enable reproducible delivery of successful configurations into cells without requiring selection, while facilitating further design refinement.

As with the first iteration of any novel approach, SGE has limitations. First, SGE requires a clearly selectable phenotype, which may not be available for many biologically interesting traits. Second, its reliance on viral integration means that, although sequencing identifies

which cassettes are present in a cell, it does not necessarily capture the expression of individual genes, which can vary with integration copy number and genomic context.

With future technological development, SGE could be applied to a wide range of biotechnology problems, such as engineering cells to synthesize a broad range of metabolites, combining multiple biosynthetic capabilities within the same cell, and reducing dependence on costly medium components during the manufacturing of viral vectors and biologics⁶. Such studies could also reveal how enzymatic burden, flux competition and cellular energetics shape performance as programs become more complex. SGE could also be applied to improve cell-based therapies by identifying metabolic and regulatory programs that help immune effector cells withstand nutrient limitation, hypoxia and suppressive signals in the tumor microenvironment⁸. Beyond metabolism, SGE could be used to discover combinations of factors driving cellular reprogramming or differentiation⁹.

The data generated from SGE screens could prove just as valuable as the engineered products themselves⁷: large, systematic genotype-to-phenotype maps generated by SGE could be used to train predictive models of mammalian metabolism. Foundation models trained on genomic and protein structure data are beginning to transform how we read and design genetic systems, but predicting how enzyme networks function as an integrated system inside cells remains a central challenge for future research¹⁰. Paired with high-throughput metabolomics, SGE data could be used to train foundation models that account for cellular metabolic fluxes and energy balance in engineered cells. If SGE fulfills this promise, it could accelerate metabolic pathway construction by closing the loop between large-scale perturbation, model training and predictive pathway design.

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